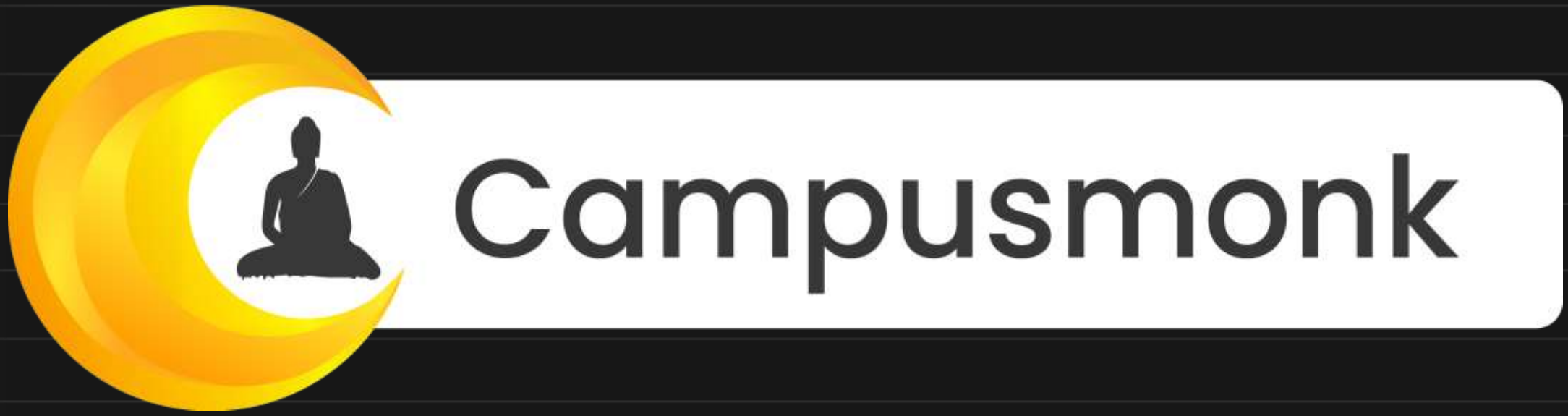

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Mensuration 3D

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1. The maximum length of a pencil that can be kept in a rectangular box of dimensions 8 cm x 6 cm x 2 cm is:

(a) $2\sqrt{13}$ cm (b) $2\sqrt{14}$ cm (c) $2\sqrt{26}$ cm (d) $10\sqrt{2}$ cm

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2. The sum of the length, breadth and height of a cuboid is 19cm and its diagonal is 5√5cm. It's surface area is : (S.S.C. CHSL 2017)

(a) 125 cm^2 (b) 236 cm^2 (c) 361 cm^2 (d) 486 cm^2

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3. The ratio of the volume of a cube to that of a sphere which will fit inside the cube is :

(S.S.C. CGL 2019)

(a) 4 : π (b) 4 : 3π (c) 6 : π (d) 2 : π

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4. How many lead shots each 3 mm in diameter can be made from a cuboid of 9 cm x 11 cm x 12 cm ?

(a) 7200 (b) 8400 (c) 72000 (d) 84000

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5. The volumes of two spheres are in the ratio of 64 : 27. The ratio of their surface area?

(a) 1 : 2 (b) 2 : 3 (c) 9 : 16 (d) 16 : 9

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6. The radii of two cylinders are in the ratio of 2 : 3 and their heights are in the ratio of 5 : 3. The ratio of their volumes is : (SSC CHSL 2018)

(a) 4 : 9 (b) 9 : 4 (c) 20 : 27 (d) 27 : 20

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7. The ratio of total surface area to lateral surface area of a cylinder whose radius 20cm and height 60cm is ?

(a) 2 : 1 (b) 3 : 2 (c) 4 : 3 (d) 5 : 3

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8. Three cubes with sides in the ratio 3 : 4 : 5 are melted to form a single cube whose diagonal is $12\sqrt{3}$ cm. The sides of the cubes are: (M.A.T. 2018)

(a) 3 cm, 4 cm, 5 cm (b) 6 cm, 8 cm, 10 cm

(c) 9 cm, 12 cm, 15 cm (d) None of these

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9. How many cubes of 10cm edge can be put in a cubical box of 1m edge?

(a) 10 (b) 100 (c) 1000 (d) 10000

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10. If the areas of three adjacent faces of a rectangular block are in the ratio of 2 : 3 : 4 and its volume is 9000 cu. Cm, then the length of the shortest side is? (TCS 2020)

(a) 10 cm (b) 15 cm (c) 20 cm (d) 30 cm

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11. What is the ratio of the volumes of a cylinder, a cone and a sphere if each has the same diameter and the same height ? (TCS 2019)

(a) 1 : 3 : 2 (b) 2 : 3 : 1 (c) 3 : 1 : 2 (d) 3 : 2 : 1

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12. 50 men took a dip in a water tank 40m long and 20m broad on a religious day. If the average displacement of water by a man is 4 m^3 , then the rise in the water level in the tank will be : (N.I.F.T. 2018)

(a) 20 cm (b) 25 cm (c) 35 cm (d) 50 cm

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13. A hall is 15 m long and 12 m broad. If the sum of the areas of the floor and the ceiling is equal to the sum of areas of the four walls, the volume of the hall is:

(a) $720m^3$ (b) $900m^3$ (c) $1200m^3$ (d) $1800m^3$

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14. 12 spheres of the same size are made from melting a solid cylinder of 16cm diameter and 2cm height. The diameter of each sphere is? (CET 2017)

(a) √3 cm (b) 2 cm (c) 3 cm (d) 4 cm

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15. The height of a wall is six times its width and the length of the wall is seven times its height. If volume of the wall be $16128m^3$, its width is: (C.B.I. 2018)

(a) 4 m (b) 4.5 m (c) 5 m (d) 6 m

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16. The edges of a cuboid are in the ratio 1:2:3 and its surface area is 88 cm^2. The volume of the cuboid is: (S.S.C. 2019)

(a) 24 cm^3 (b) 48 cm^3 (c) 64 cm^3 (d) 120 cm^3

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17. Three cubes of metal of edges 6 cm, 8 cm and 10 cm respectively are melted and recast into a single cube. The edge of the new cube formed is: (TCS 2018)

(a) 12 cm (b) 14 cm (c) 16 cm (d) 18 cm

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18. The curved surface of a right circular cone of height 15cm and base diameter 16cm is : (H.Mgmt. 2019)

(a) 60π cm^2 (b) 68π cm^2 (c) 120π cm^2 (d) 136π cm^2

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19. If the curved surface area of a right circular cone is 3080 cm^2 and its radius is 28 cm, find its volume?

(1) 8624cm^3 (2) 4312cm^3 (3) 17248cm^3 (4) 34496cm^3

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20. A cylindrical vessel of radius 7cm is partially filled with water. By how much will the water level rise if a sphere of radius 6cm is completely immersed in this water?

(1) 11.76cm (2) 17.64cm (3) 23.52cm (4) 5.88cm

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21. Water flows at the rate of 10m/sec from a cylindrical pipe 5mm in diameter. How long it will take to fill up a conical vessel whose diameter at the base is 30cm and depth 24cm?

(1) 28min 48sec (2) 51min 12sec (3) 51min 24sec (4) 28min 36sec

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22. A copper sphere of diameter 18cm is drawn into a wire of diameter 4mm. The length of the wire, in metres is? (M.A.T. 2017)

(1) 2.43 (2) 243 (3) 2430 (4) 24.3

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23. A cone and a hemisphere stand on equal base and have the same height. The ratio of their whole surface areas is:

(1) (√2 + 1):3 (2) (√2 - 1):3 (3) √2:3 (4) 2:3

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24. If radius and height of a cylinder are increased by 20% each then by how much percent the volume of the cylinder will be increased? (S.S.C. 2020)

(1) 68.2% (2) 72.8% (3) 66.8% (4) 78.2%

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25. Find the ratio of the curved surface area and volume of a right circular cone of radius 12cm and height 5cm: (S.S.C. 2019)

(1) 14:23 (2) 20:13 (3) 13:20 (4) 23:14

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26. Three cubes each of volume $216m^3$ are put together in such a way that each is placed on the top of the other forming a large cuboid. Find the total surface area of the cuboid?

(1) $432m^2$ (2) $216m^2$ (3) $504m^2$ (4) $698m^2$

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27. Find the volume of water fall on a 1.5Hectare area of land if there was a 5cm rainfall on a particular day? (N.I.F.T. 2017)

(1) $75m^3$ (2) $750m^3$ (3) $7500m^3$ (4) $75000m^3$

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